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Course Code : EE 4121

VLSI

Introduction to VLSI Technology

Syllabus

- Introduction to microelectronics and MOS technology
- Basic electrical properties and circuit design processes of MOS and Bi-CMOS circuits
- Layout design rules, sub-system design processes
- Computational elements: design of CPU and ALU sub-system, data path, adder, multipliers, memory, registers, and aspects of system timing, practical aspects of design tools and testability
- Device modeling: MOS, diode, bipolar and passive component
- CMOS Design: Behavioral description, structural description, physical description and design verification.

Reference Books:

1. **Basic VLSI Design by Douglas A. Pucknell**

2. **Design of VLSI Systems-A Practical Introduction by Linda E.M. Brackendary**

Introduction to Microelectronics

An **integrated circuit (IC)** is a compact electronic device in which hundreds to millions of interconnected transistors, resistors, and capacitors are fabricated on a thin semiconductor substrate, typically silicon, to create a functional chip or wafer.

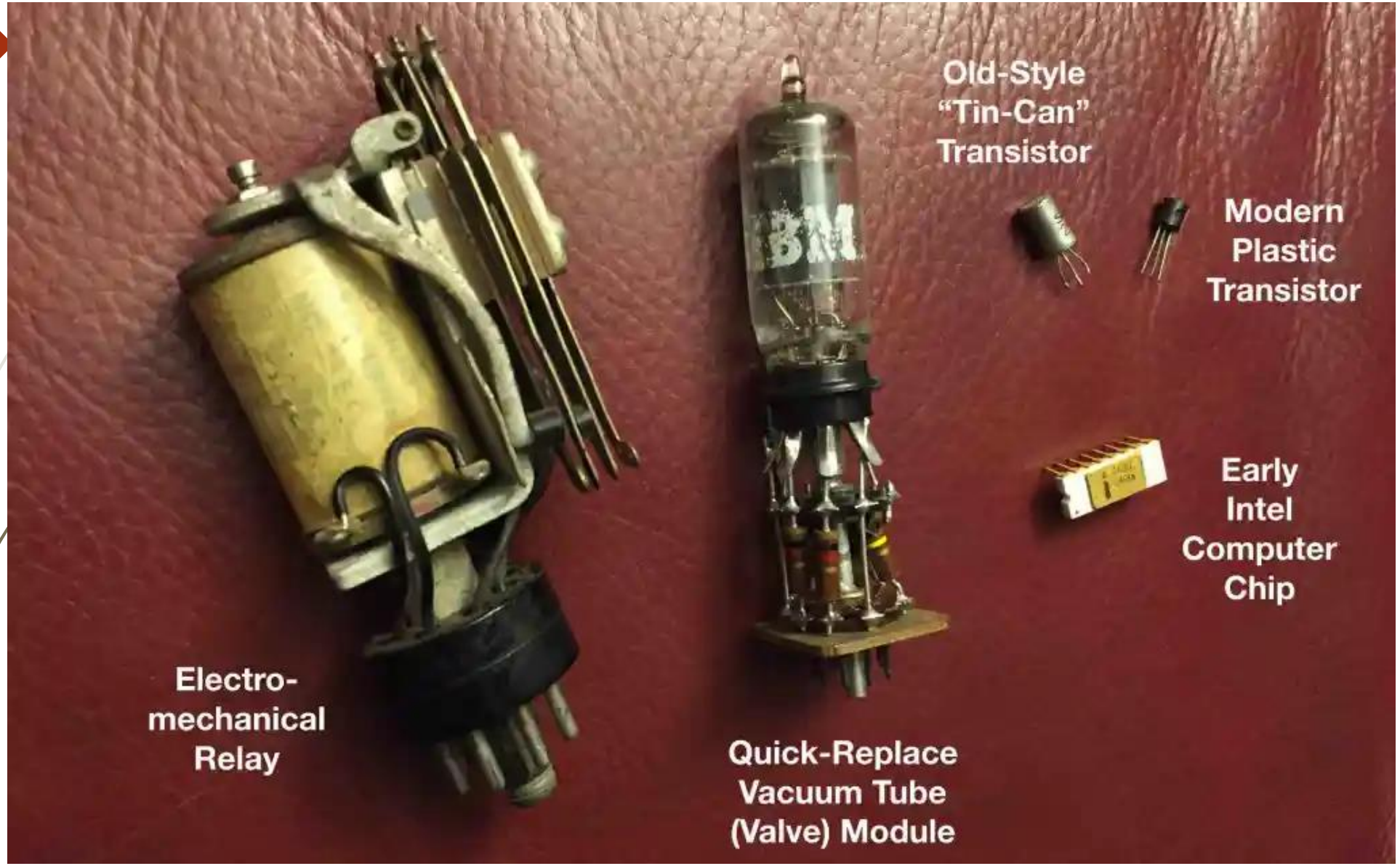
- SSI - Small Scale Integration
- MSI - Medium Scale Integration
- LSI - Large Scale Integration
- VLSI - Very Large Scale Integration
- ULSI - Ultra Large Scale Integration
- GLSI - Giant Large Scale Integration



A Brief History of Microelectronics' Evolution

TABLE 1.1 Microelectronics evolution

Year	*1947*	*1950*	*1961*	*1966*	*1971*	*1980*	*1990*	*2000*
Technology	*Invention of the transistor*	*Discrete components*	*SSI*	*MSI*	*LSI*	*VLSI*	*ULSI**	*GSI†*
Approximate numbers of transistors per chip in commercial products	1	1	10	100–1000	1000–20,000	20,000–1,000,000	1,000,000 10,000,000	>10,000,000
Typical products	—	Junction Transistor and diode	Planar devices Logic gates Flip-flops	Counters Multiplexers Adders	8 bit micro-processors ROM RAM	16 and 32 bit micro-processors Sophisticated peripherals GHM Dram	Special processors, Virtual reality machines, smart sensors	


Old-Style
"Tin-Can"
Transistor
Modern
Plastic
Transistor
Early
Intel
Computer
Chip
Electro-
mechanical
Relay
Quick-Replace
Vacuum Tube
(Valve) Module

Why VLSI/Miniaturization required in electronics?

- low size, weight and volume
- low cost (because cost depends upon size, manufacturing cost, maintenance cost etc.)
- reduced design time and complexity (realization of complex circuits using discrete devices are difficult and costly too)
- higher speed
- high degree of reliability, repeatability
- lower static power dissipation and sophisticated designs can be made

Moore's Law

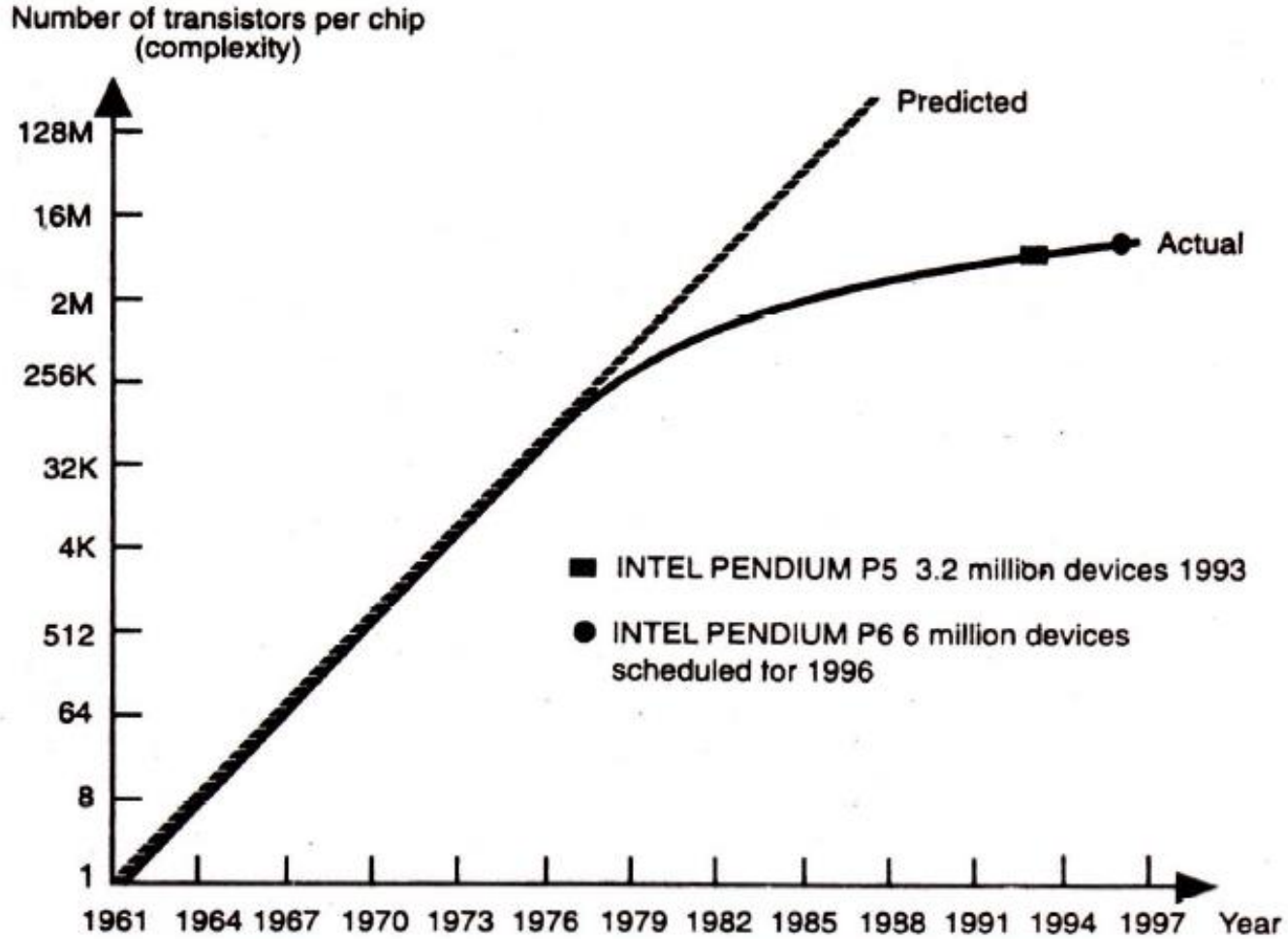
In VLSI, **Moore's Law** is the observation made by Gordon E. Moore (co-founder of Intel)

❑ 1st Law:

- The number of transistors per chip roughly doubles in every two years.
- The number of transistors per chip would grow exponentially.

❑ 2nd Law:

- The cost of a semiconductor chip fabrication doubles every four years.
- The cost of a semiconductor chip fabrication increases exponentially over time.



Note: K signifies a multiplier of 1024 and M a multiplier of 1,048,576.

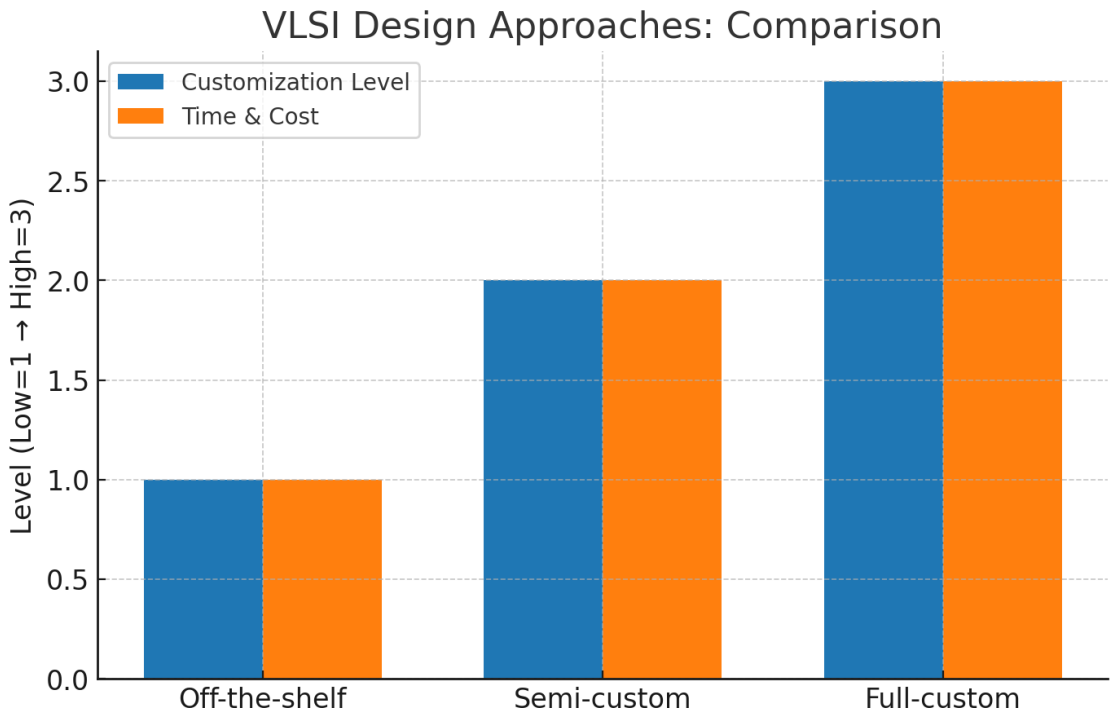
FIGURE 1.1 Moore's first law: Transistors integrated on a single chip (commercial products).

VLSI Design Approaches

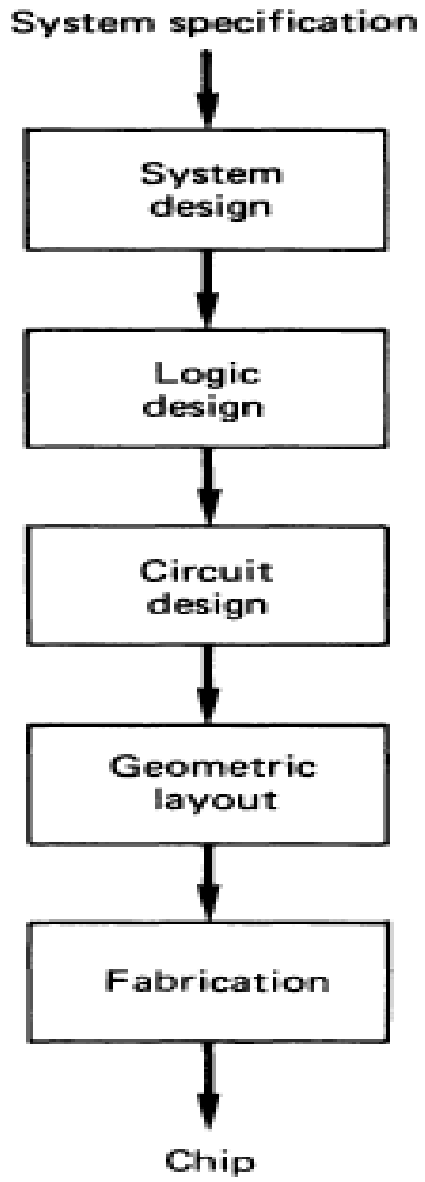
There are three approaches to implementing digital design:

- 1st Approach – Off the self. The chips are available " off the self" from manufacturers.
  - designer is limited to the integrated circuits on offer
  - availability of specific functions influences the resulting design
- 2nd Approach – Full custom design. The chip design is completely depends on designer.
  - total control over the chip function including the specification of the content of each layer manufactured
  - associated overheads of the design time, development and manufacturing cost
- 3rd Approach – Semi custom design.
  - design in the form of the uncommitted logic array (ULA)
  - designer specify the cell interconnection and the cell configuration to provide a user-specified function

Approach	Level of Customization	Speed to Market	Performance Optimization	Cost
Off-the-shelf	None	Fast	Fixed by vendor	Low
Full-custom	Maximum	Slow	Best possible	High
Semi-custom	Moderate	Medium	Good	Medium



Full Custom Design/ Top-down Design Hierarchy



Choice of VLSI Technology

- Choice of materials:
  - Low cost
  - Availability
  - Insulating property

Si/SoS/Ge/GaAs

- Why Si is better than others?/ Why Si technology is most popular?
  - GeO_2 – is partially soluble
  - GaAs – does not form a oxide, have higher mobility but high integration levels are difficult to achieve